An Augmented Reality Application for Mobile Visualization of GIS-Referenced Landscape Planning Projects

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Figure 1: Several virtual objects, which are represented with our application with occlusion

ABSTRACT

We introduce an augmented reality application that allows the representation of planned real world objects (e.g. wind turbines or power poles) at their actual geographic location. The application uses GPS for positioning, which is then supplemented by augmented reality feature tracking to get a constant and stable positional and rotational reading. As addition to the visualization, we use on-the-fly gathered sensor data to identify foreground objects. Thus, for practical scenarios our application depicts images with mostly correct occlusion between real and virtual objects. The application will be used to support urban and landscape planners in their process, especially for the purpose of public information and acceptance. It provides an advantage to current planning processes, where the representation of objects is limited to positions on maps, miniature models, or at best a photo montage where the object is placed into a still camera image.

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CCS CONCEPTS

• **Information systems** → **Geographic information systems**; • **Human-centered computing** → *Geographic visualization*; • **Computing methodologies** → *Mixed / augmented reality*.

KEYWORDS

GIS, augmented reality, visualization

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1 INTRODUCTION

The planning of large structure projects such as wind turbines or power lines is currently a large challenge. Public acceptance is a very important factor in the planning process, as low acceptance of a planned project can lead to disputes and lawsuits which often result in long delays and changes to plans. For example, about 20 percent of currently approved wind power projects in Germany are subject to lawsuits, some of them spanning years [Fachagentur Windenergie an Land 2019]. Most of them are done by nature conservation organisations for protection of endangered species,

but there are also many that are disputed by local residents or citizen initiative groups. Accurate visualization of those wind turbines can help planners convey to locals and authorities the impact on the landscape. This can already be done using photo compositions of the structures from various angles, but making those for a lot of different views can be time consuming and potentially expensive.

We are in the process of developing an augmented reality application that allows one to visualize such planned objects in real time. It can be used to bridge the planning processes using GIS (geographic information system) applications and the demand and need for visualizations for public acceptance. The great advantage of GIS - and why it is used - is that it can be used to check for zoning restrictions or nearby structures, as seen for example in [Harper et al. 2017]. After using this process to decide on a desirable place for an object, our application takes this data directly from the GIS. Using a 3D engine for visualization allows for a multitude of features such as changing details on the object, configuring lighting, and in the example of wind turbines change orientation and animating them. We use Unity3D for this purpose. Finally, current state of the art augmented reality technology makes all this usable in real time on mobile devices. For cross-platform compatibility on Android and iOS devices, AR Foundation is used. It is Unity3D's common AR interface which manages the native ARCore (Google/Android) and ARKit (Apple/iOS) APIs [Miller et al. 2018]. This results in an application that is a powerful visualization tool for planners, authorities, and the general public.

In the following sections we will first look at related work and studies about use cases of 3D visualization for GIS systems for planning purposes as well as location based augmented reality content. Then we detail the interconnection between GIS, 3D and AR in our application, as well as various important challenges and considerations we have faced in development.

2 RELATED WORK

Example studies for augmenting GIS information with 3D data for the purpose of supporting urban or landscape planners have been evaluated by various authors. For example, in a paper by [Chundeli 2017], the planned development of an urban area is compared to the actual development. Visualized 3D content makes a mismatch between the two, and numerous zoning regulation violations very clear, and it is concluded that this kind of content can help planners avoid this. As another example, [Amirebrahimi et al. 2016] use 3D building model data for simulating flood risks. These examples show how 3D visualization can help those involved in urban and landscape planning.

The idea of location based objects in mobile outdoor augmented reality has already been analysed and also realized in multiple other projects. In a paper by [Schmid and Langerenken 2014] the possibility and accuracy of location based mobile augmented reality content is evaluated and found to be dependent on the accuracy of the devices' GPS readings which may be good enough for some types of applications but not for others. Tricks like aligning virtual objects with corresponding real objects may be used to improve accuracy when available, while for example a pure visualization of non-visible content like underground pipes is seen as problematic [Schmid and Langerenken 2014].

A common use for location based augmented reality content, especially on mobile platforms, appears to be tourism oriented applications. Including AR content in tourism related applications can enhance the user's experience according to a study by [Lee et al. 2012]. As a recent example, [Panou et al. 2018] have used it for displaying past states of historical buildings for the purpose of a mobile tourist guide.

3 SYSTEM OVERVIEW

Our system consists of several components. On the one hand the end application on the smartphone and on the other hand the backend, which provides various data and functions. These are only made available after a successful authentication. Figure 2 shows an overview of the components.

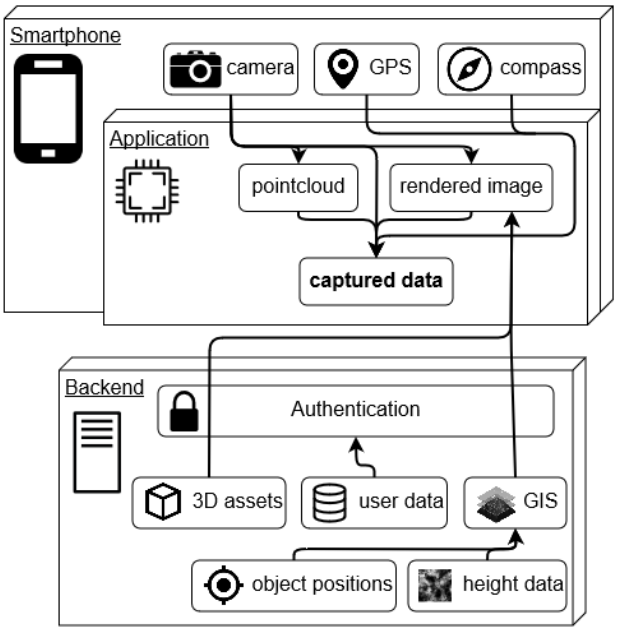


Figure 2: Component overview

The application uses several sensors on the smartphone, in particular the camera, the GPS sensor and the compass but also the accelerometer, the gyroscope and the magnetometer. The camera image is then used to generate the point cloud and to render an augmented reality picture in real time. Furthermore, 3D objects are displayed based on the data delivered by the backend. The GIS server provides height maps and positions of the objects. The corresponding 3D models are also obtained from the server. Depending on the application more or less 3D models are downloaded. Thus the actual application can be kept light and flexible. The user can capture the data for a specific moment. The captured data includes the generated point cloud, a rendered augmented reality image, the pure camera image, the pure virtual image with alpha channel and metadata like the GPS position and the compass. The data makes it possible to edit the image afterwards and even to generate a completely new image.

4 GIS INTEGRATION

GIS systems are typically the tools in use for landscape planning purposes. They can work with and reference map data and databases with locations. Our goal for the application is to interoperate with these systems for a convenient workflow.

Batches of information in GIS systems are typically split into different layers.The application makes use of three layer types, which are loaded via web service from our GIS server. The most important layer type is the feature layer, which contains features, i.e. planned objects at their positions. It is essentially a database that is accessed through the GIS system. We use a WFS (Web Feature Service) to access it which is specified by the Open Geospatial Consortium [Consortium 2019]. For organizational purposes, our application loads projects which can each have multiple of these layers. Additional map data is provided in form of a height map texture for a chosen area. This is used to position the user and objects on the height axis and is provided using a WCS (web coverage service). Finally, we cache some satellite map data of a project area to help the user's orientation and allow in-field adjustments. A WMS (web map service) is used to access this data.

The feature database does not directly contain 3D model data, this has to be prepared separately as Unity3D-compatible "asset bundles" [Foundation 2019]. The advantage is easy model replacement, for example visualizing different types of wind turbine on the same position. These are possible to download and integrate at runtime, so on the server side there is not only the WFS service with the model descriptions and locations but also a directory of model data that is referenced by the model descriptions. The model data itself has to be processed by Unity3D for creating the asset bundle but can be originally sourced from most common 3D formats.

With this setup it is possible to prepare map data and feature data using GIS, and then use our application to visualize it. Moreover it is also possible to relocate features in field use. While our application does not respect zoning laws and allowed areas as GIS tools can, we can write back to the WFS layer to review changes later on back in the GIS workflow.

5 VISUALIZATION

A key aspect of visualization is that the virtual object is inserted at the right depth. The only real world depth information available to us is currently a sparse point cloud of image features and their positions that is available through ARFoundation [Miller et al. 2018]. It uses ARCore's and ARKit's respective functionality.

In our current use case with the example of wind turbines - which are large structures at a large distance away - the depth problem can usually be greatly simplified. In many cases everything that is not sky is in front of the turbine, and the sky is behind it. Our solution for this case is a one-time manual selection of an area in the image which is sky. From this selection, a small selection of sky colors is extracted by means of simple k-means-clustering of the pixel colors after transforming them to a suitable HSV hexcone representation [Smith 1978]. Using a threshold from these colors, everything in the image can be marked as sky or not sky and put in front of or behind virtual objects.

Beyond that, an attempt has been made to use the point cloud functionality to extract a depth for the whole image using an image pyramid based method, as described in section 6 in [Strengert et al. 2006]. This experiment yielded mixed results. It is reliant on typical image features; strong edges or corners. These are not always present in an image. The way they are derived is also not openly disclosed in the API, so there is no way to finely configure the point cloud output or to specify regions of interest in the image for feature search. It seems to be working quite well for edges of close objects, which does make sense because after all the features are likely derived in a SLAM (simultaneous localization and mapping) method for AR device localization which operates on image parts "moving" from one camera image to the next. However on further away objects, not as many feature points are found. A method that operates only on point cloud data will not be reliable in these cases, something that incorporates the camera image will have to be used. Research on something like this has been done already in [Cruz-Martinez et al. 2018], but we have not tried to integrate this into our application yet.

Another visualization aspect is proper lighting. The goal for the best representation is to light the virtual objects in a way that it equals the way real objects would be lit. While a modern graphics engine such as Unity3D is capable of rendering illuminated 3D objects quite realistically with the right parameters, real outdoor lighting conditions however can vary greatly based on time, location and weather. As we do know the first two of these parameters, time and location, we can calculate the sun position and thus the illumination angle. This can be used for accurate shadows as well. Weather however can change the lighting depending on if it is clear weather, cloudy, foggy or raining. The same object in different conditions can look very different or might not even be visible at all. Here the user has to adjust certain lighting parameters for themselves. The user adjustable parameters in our application are lighting intensity and temperature as well as fog density and brightness. With these a large spectrum of conditions are possible to simulate.

All these visualization efforts are great for displaying things in the AR application, but sometimes a single captured photo is desired. Additionally, sometimes the application does not display perfect results. In some cases the foreground/background model is wrong, or not all of the sky is correctly recognized. For this there is the functionality to take a capture of the current display. It does not only take a picture of the screen, but also saves depth, sky-mask and separate layers of camera image and virtual-objects-only image. From all this image information it is possible to adjust details after taking a capture. An example is visible in figure 3 which shows camera image, sky mask and virtual objects for our teaser image.

From these pieces, the whole image can be precisely reconstructed while also altering things. For example wrongly classified areas of sky/non-sky can be fixed so the virtual objects are displayed correctly. This feature makes the application not only provide live visualization but also powerful "post-production" possibilities.

6 ACCURACY

The quality of the visualization depends on the accuracy of the virtual camera position and orientation. These can be determined from a combination of the device's sensors. This is where we ran into challenges.



Figure 3: Different layers of a captured image. From left to right: Camera image, sky-mask, and rendered image

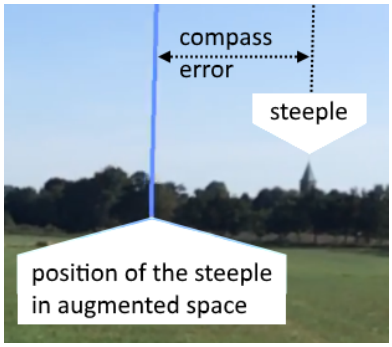


Figure 4: Compass rotation error in an example situation

The original plan for the application was to use the compass, the GPS data, the accelerometer and the gyroscope to accurately locate the user and match the location and orientation with the virtual world. For preliminary testing of our application, we did place objects at the positions of a couple well visible landmarks and then noticed that their location in the AR image was noticeably off. From logged position/rotation data and comparison with map data we determined that the error was in the device's sensor readings. The GPS sensor is inaccurate on the scale of a few metres, which has also been found by other researchers while working on their projects [Schmid and Langerenken 2014][Erifiu and Ostermayer 2011]. Additionally, since we are looking at objects that are very far away, inaccuracies in the compass heading cause large positional errors of virtual objects.

Finally, not only the GPS latitude/longitude position is inaccurate, but also its elevation or height value. This can also affect the displayed objects. Small inaccuracies in position and alignment can add up to large errors in the visualization. For most applications, these errors are tolerable, but great accuracy is required when planning infrastructure. The conclusion here is that these errors must be dealt with. On our target systems of handheld mobile devices replacing the sensors by more accurate ones is not possible, so improvements would have to come from workarounds, alternative data, or manual user entry.

As a result, we decided that the application needs a way to manually adjust position, orientation and height to be able to match the virtual world to the real one. For position, an accurate map where a user can select their position can accomplish this. For rotation our approach is to use decently visible landmarks after position is fixed. As we can already position AR content on accurate geographic positions, we can put a marker on a landmark which can be chosen in preparation or in the field on the satellite image map. So after fixing the position, the user can calibrate the rotation so that the marker lines up with the real world object.

Figure 4 shows an example of the rotation step, where a virtual line is displayed where the steeple should be. Due to the aforementioned compass heading inaccuracy, there is a difference between the virtual and real world. Rotating the virtual world so the line ends up on top of the real object fixes this.

Since there is terrain height data available from the map server, our workaround for height is to place the user a default height above the terrain elevation. An option to adjust the height from there is available as well, in case of use from an elevated position such as a building. After adjusting according to these three steps, the virtual world is properly adjusted so it matches the real world.

7 CONCLUSION

While the application is still under development, it has been well received in initial tests with potential end users. Checking and, if necessary, correcting the position and alignment does indeed mean additional work for the user. As such the usage of the application is not as easy as imagined at first. However, due to the inaccuracies of the sensors, there is no other way to ensure that the visualization is correct. While this ultimately does leave room for user error, the same potential of user error also applies to photo compositions or other visualizations. The augmented reality application, however, makes the visualization process easier and faster.

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